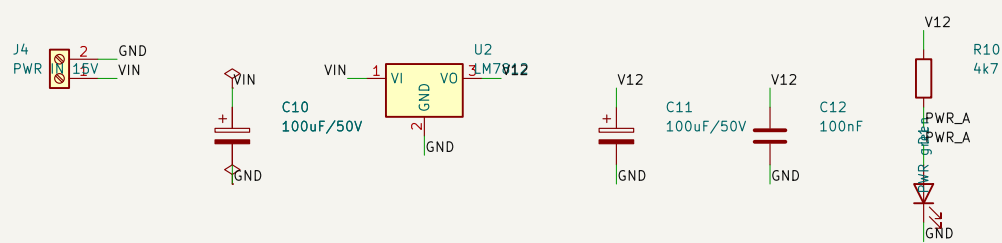


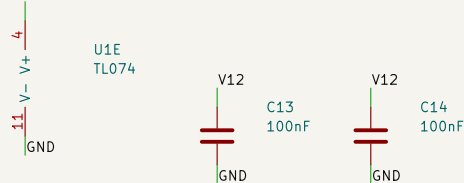
Bose Companion 5 sub crossover -- mono-summing 2nd-order Sallen-Key low-pass.

Single +12 V rail, no negative supply. All AC referenced to VGND = 6 V (A3 buffers the R8/R9 mid-rail divider).

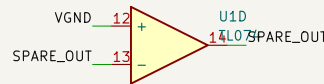
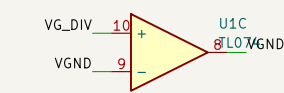
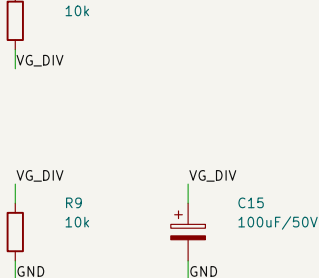
A POWER IN $\rightarrow +12\text{ V}$



B U1 SUPPLY



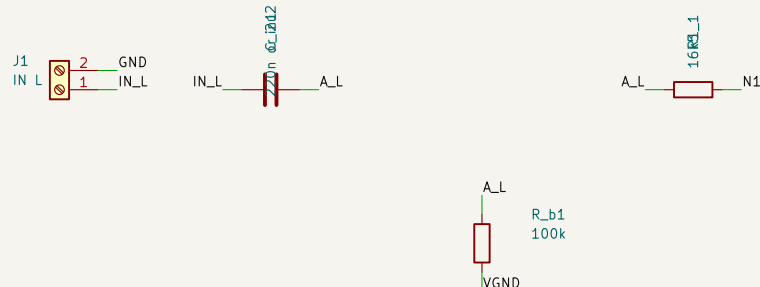
C VIRTUAL GROUND 6 V



U1 supply: pin 4 = V12, pin 11 = GND. No negative rail.

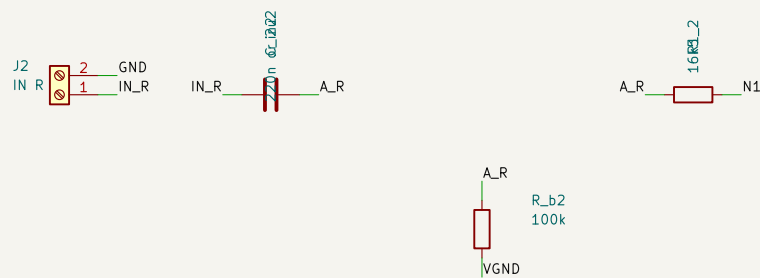
A4 is the spare section: + on VGND, out tied to in-.
An unterminated spare oscillates and couples into its neighbours through the shared supply.

D INPUT L

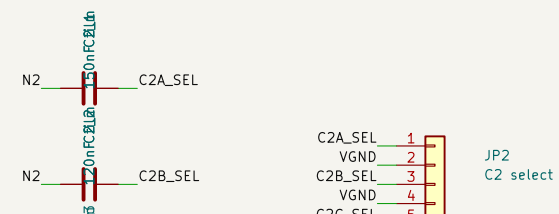
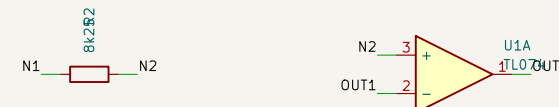
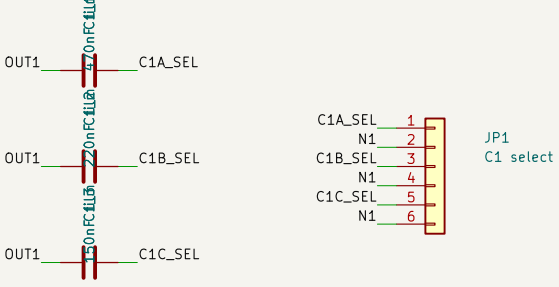


R_b1/R_b2 are the only DC path to A1's + input (C_in blocks DC, C2 is a capacitor). Without them A1 drifts to a rail. They sit after the coupling caps -- 100k at N1 would shift every corner by 8%.

E INPUT R



F Sallen-Key Low-Pass



C1 returns to N1 -- the Sallen-Key feedback path. Not N2.
JP1 selects C1 (odd pin = cap, even pin = N1); JP2 selects C2 (odd = cap, even = GND). They select INDEPENDENTLY -- nine corners, not three pairs. Corners shown are MEASURED on the rev A board, not calculated: the quoted $f_0 = 1/(2\pi R \sqrt{C_1 C_2})$ is not a -3 dB point, and C_{in} sits inside the filter. See Results -- Sub Crossover Bring-up.

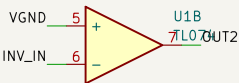
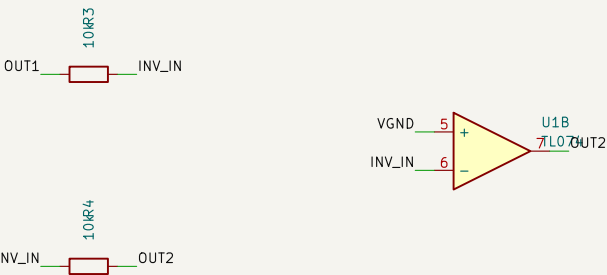
Intended use is a pos 1 2-pole 3-position rotary, one pole per header:

pos 1 94 Hz	A0→JP1 even, A1→JP1.1 (470n)	B0→JP2 even, B1→JP2.1
pos 2 136 Hz	A2→JP1.5 (150n)	B2→JP2.3
pos 3 189 Hz	A3→JP1.3 (220n)	B3→JP2.5

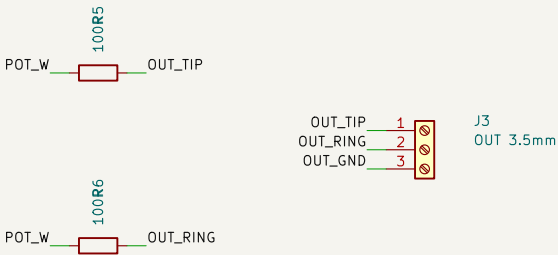
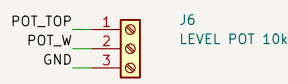
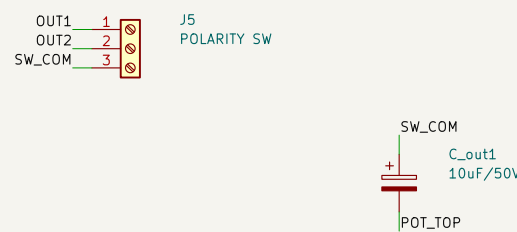
NEVER wire one header pin to two switch lugs unless both lugs want the same contact. The A0's are then tied together and every position collapses to the same value. This is why C1.1 is fitted rather than paralleling C1.2 | C1.3 on one lug.

Shunts across a pin pair still work as a bench fallback.

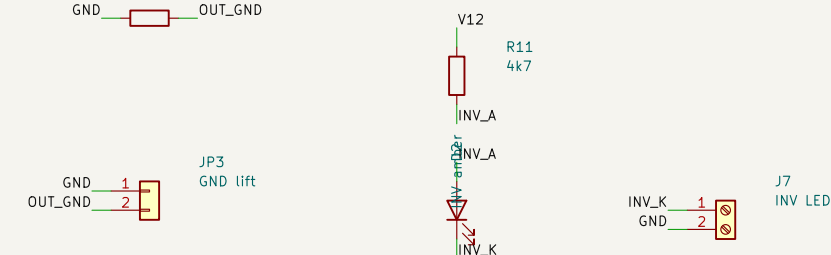
G POLARITY INVERTER



H OUTPUT STAGE



I STATUS LEDS



C_{out} + faces SW_{COM}: that node sits at 6 V DC, the pot side at 0 V.
 J6 pin 3 = GND, not VGND — C_{out} has already removed the 6 V bias;
 referencing the pot to VGND would thump the driver at power-on.
 JP3 shorts R7 (ground lift). Normally fitted.

D1 = power lamp on V12 (proves the LM7812, not just VIN).
D2 = INVERTED lamp, fed from SW2's unused second pole via J7.
Both ~2 mA. No LED touches a signal node.

Off-board: level pot (#6), polarity switch (#5), all panel jacks (#12/#12/J3).
 Process: single-sided THT, bottom copper only, board outline 100 x 100 mm.
 Copper is isolation-milled on the CNC with an 0.8 mm flat end mill; the fiber
 laser does the silkscreen only. Routing clearance 0.85 mm / track 1.0 mm.
 The 0.8 mm tool width is a hard floor on every copper-to-copper gap. Fixed-pitch
 pads sit below the 0.85 mm target and carry a .kicad_dru exception: DIP-14
 0.94 mm (passes), 2.54 mm headers and the T0-220 0.84 mm (do not).
 Drills: 0.8 mm x52, 1.0 x26, 1.1 x3, 1.2 x4, 1.5 x15.

Reference names vs the handoff document: KiCad treats a reference with no trailing digit as unannotated, so C1a/C1b/C1c -> C1_1/C1_2/C1_3, C2a/C2b/C2c -> C2_1/C2_2/C2_3, R1a/R1b -> R1_1/R1_2, C_out -> C_out1.

Single-sided THT, CNC isolation milling (0.8 mm end mill), laser silkscreen
Mono-summing 2nd-order Sallen-Key low-pass, single +12 V rail

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Title: Bose Companion 5 sub crossover

Size: A2	Date: 2026-08-16	Rev: B
KiCad: EDA 10.0.4		Id: 1/1